Evaluating U.S. HDPE Circularity Through Quality-Resolved Dynamic Material Flow Analysis

Supplementary Methodology and Model Parameterization

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This document provides supplementary methodology, model parameters, input data, mathematical formulations, and supporting tables for the accompanying MSEC 2027 paper. Sections S1–S9 describe the model assumptions and implementation, while Appendix A provides additional reference data and allocation matrices.

S1. Baseline U.S. HDPE flows and model nodes

The 2015 baseline structure is anchored to the U.S. HDPE MFA of Chatterjee et al. [1] . The original flow matrix is retained as a reference and as the basis for selected derived quantities, including 2015 domestic product demand, finished-product import/export ratios, the status-quo collection fraction, and the residual landfill-to-incineration split. In the current model, the historical cohort model rather than the U1/U2/U3 stock inputs determines stock and EoU generation. Table S 1 highlights the values used from the 2015 static MFA [1] and the purpose of their use in the model.

Table S 1. 2015 baseline HDPE flow matrix (original units: kt).

Origin	V	RC	F	U1	U2	U3	EoU	Inc	Land	Export	Current use in the present model
V — Virgin resin production	0	0	**7,643**	0	0	0	0	0	0	**916**	The 2015 virgin-resin flows are retained as baseline/reference information. Virgin production is **not fixed at 7,643 kt** in the current model; annual virgin production is demand-responsive after feasible recycled resin is allocated. The historical virgin-export flow is used to establish the normal virgin-export relationship to domestic product demand.
RC — Recycled resin	0	0	**563**	0	0	0	0	0	0	0	The 563 kt recycled-resin flow is **not imposed as a fixed recycled-content input** in the current model. Recycled-resin production and use are determined annually from collection, sorting, recycling yields, MFI compatibility, and optimization.
F — Fabrication and manufacturing	0	0	0	**3,661**	**1,748**	**1,209**	0	0	**1,618**	**42**	The three product-output flows provide the 2015 domestic product-production reference, 3661+1748+1209=6618 kt. The 42 kt product export is used to derive the baseline finished-product export ratio. The 1,618 kt fabrication loss is retained in the fabrication-yield/material-efficiency basis rather than imposed as a fixed annual waste flow.
U1 — Packaging and containers	0	0	0	0	0	0	**≈3,429**	0	0	0	The original U1→EoU value is no longer imposed. Current EoU generation is calculated dynamically from historical application×sector cohorts and lifespan distributions.
U2 — Durable goods	0	0	0	0	0	0	**≈1,343**	0	0	0	Retained only as a historical/reference flow. Current durable-product retirements are generated from the cohort model rather than from this fixed 2015 arrow.
U3 — Non-durable goods	0	0	0	0	0	0	**≈708**	0	0	0	Retained only as a historical/reference flow. Current EoU generation is cohort- and lifespan-based.
EoU — End-of-use HDPE	0	**563**	0	0	0	0	0	**862**	**4,055**	0	The historical recycling flow is used to define the **status-quo collection/recovery rate for R1**. The landfill and incineration values are used to derive the **normalized residual disposal split** applied to uncollected EoU material. The absolute 2015 EoU quantities are not forced in the dynamic simulation.
Inc — Incineration	0	0	0	0	0	0	0	0	0	0	Terminal sink; the original absolute mass is not imposed annually.
Land — Landfill	0	0	0	0	0	0	0	0	0	0	Terminal sink; current landfill receives uncollected EoU, unused collected HDPE, fabrication losses, and recycling/sorting losses as applicable.
Import	0	0	0	0	**4**	**37**	0	0	0	0	The 41 kt total finished-product imports are used to derive the baseline product-import ratio. Current product imports scale with modeled domestic product demand. Virgin-resin imports for fabrication are disabled in the reported analysis.

S2. Historical (1950—2018) cohort initialization, in-use stock, and EoU generation

The HDPE use-sector shares shown in Table S 2 are used for modeling U.S. HDPE consumption for years 1950-2018. The table also lists the lifespan of HDPE used in each of these use-sectors. These mean lifespans and standard deviation values are used for the historical cohorts 1950-2018 and modern HDPE 2019-2060 [2,3].

Table S 2. Use-sector shares and lifespan distributions.

Sector_ID	Historical HDPE Share	Lifespan Mean (year)	Lifespan SD (year)	Lifespan Distribution	Active Through Year	Notes
Packaging	0.570552	0.5	0.1	lognormal	2018	Geyer/Kan; historical sector allocation through 2018 [2,3]
Transportation	0.0490798	13	3	lognormal	2018	Geyer/Kan; historical sector allocation through 2018
Building_Construction	0.202454	35	7	lognormal	2018	Geyer/Kan; historical sector allocation through 2018
Electrical_Electronic	0.0122699	8	2	lognormal	2018	Geyer/Kan; historical sector allocation through 2018
Consumer_Institutional	0.104294	3	1	lognormal	2018	Geyer/Kan; historical sector allocation through 2018
Industrial_Machinery	0.00613497	20	3	lognormal	2018	Geyer/Kan; historical sector allocation through 2018
Other	0.0552147	5	1.5	lognormal	2018	Geyer/Kan; historical sector allocation through 2018
Automotive		13	3	lognormal		2019+ modern sector; Mapped to Transportation
Consumer_Goods		3	1	lognormal		2019+ modern sector; Mapped to Consumer/Institutional
Industrial		20	3	lognormal		2019+ modern sector; Mapped to Industrial Machinery

S3. Application-grade demand and Melt Flow Index representation

HDPE demand is represented by ten explicit application/Melt Flow Index(MFI) grades. The original market share by application values is extracted from Prismane HDPE market report [4]. The commercial 'Other' category is not treated as a physical MFI grade; its reported share is redistributed proportionally among explicit grades. Combined Film & Sheet demand is split equally between film and sheet, and Pipe & Extrusion demand is split equally between pipe and thin-tube/profile extrusion. These are author disaggregation assumptions distinct from the literature-derived MFI ranges.

Each grade is assigned a literature-based MFI interval and a deterministic representative value from reference [5] . The deterministic value is the midpoint of the range unless the source provides an approximate value requiring a stated convention. Application MFI uncertainty can be sampled from the indicated distribution.

Table S 3 summarizes the by application demand shares and the minimum and maximum values that were used to establish the midpoint representing the application MFI grade.

Table S 3. Application demand shares and MFI inputs.

Application ID	Application Name	Base Demand Share pct	MFI Min g/10min	MFI Max g/10min	Virgin MFI g/10min	MFI Uncertainty Distribution	Notes
Injection	Injection molding	26.5957	1	2	1.5	uniform	Direct published range; Prismane 25% known-category share rescaled after removing Others.
Film	Film	21.2766	9	15	12	uniform	Half of Prismane Film & Sheet (author 50/50 split).
Sheet	Thin sheet	21.2766	0.5	1	0.75	uniform	Half of Prismane Film & Sheet (author 50/50 split).
Blow	Bottle / blow molding	9.57447	0.1	0.5	0.3	uniform	Bottle-grade MFI range used as blow-molding proxy.
Pipe	Pipe	6.38298	0	0.1	0.05	uniform_open	Source gives <0.1; operationalized as open interval (0,0.1).
Tube_Profile	Thin tube / profile extrusion	6.38298	0.1	0.5	0.3	uniform	Thin-tube range; replaces unsupported GenExtr. Pipe & Extrusion split 50/50 by author assumption.
Fiber	Multifilament / fiber	3.19149	0.9	1.1	1	uniform	Source ≈1.0; ±10% author uncertainty.
Wire_Cable	Wire & cable	2.12766	0.1	1	0.55	uniform	Direct published range.
Raffia	Raffia / monofilament rope	2.12766	0.5	1	0.75	uniform	Monofilament/rope range used as proxy.
Rotomold	Rotomolding	1.06383	2	8	5	uniform	Peer-reviewed 2–8 range; midpoint deterministic.

Base demand shares sum to 100% after proportional redistribution of the commercial 'Other' category.

S4. Application-to-sector allocation

Available market data provide application-grade and use-sector distributions mainly as separate marginals rather than a directly observed joint distribution. A joint application-sector matrix Jg,s is therefore constructed using iterative proportional fitting (IPF) [6] . A taxonomy-informed prior matrix is assigned weights of 0, 1, 2, or 3 to represent structurally excluded, residual-plausible, plausible/secondary, and strongly supported application-sector combinations, respectively. The prior is informed by the detailed product taxonomy of Liu and Nowack [7], but no Swiss absolute masses are transferred to the U.S. model.

IPF rescales the prior until row totals reproduce the selected application shares and column totals reproduce the sector shares. The resulting joint cells are therefore derived allocation coefficients, not measured U.S. market shares. Historical Geyer/Kan [2,3] sector composition is used through 2018. Beginning in 2019, a modern sector composition derived from digitized Prismane [4] HDPE end-use data is applied. The current implementation uses an abrupt 2019 transition; this transition is identified as a structural sensitivity.

Table S 4 lists the mean by use-sector HDPE share as extracted from Prismane HDPE market report. The full joint matrix used in the model is reported in Table S 10 and the seed matrices for fitting the joint matrix for historical use-sector and modern use-sector as well as the justification for the seed value are listed in Table S 11 and Table S 12 respectively.

Table S 4. Modern HDPE sector shares used from 2019 onward.

Sector_ID	Mean_Share_pct	Dirichlet_Alpha
Packaging	22.44	2,052.2
Building_Construction	14.11	1,290.015
Automotive	12.53	1,144.947
Consumer_Goods	14.05	1,285.352
Industrial	22.87	2,090.184
Other	14	1,280.655

Prismane digitized 2018–2034 mean. Dirichlet alpha fitted by maximum-likelihood to the 17 annual digitized composition vectors; fitted uncertainty model, not reported by Prismane.

S5. MFI mixing, compatibility, and quality memory

The MFI of fabricated blends is propagated through the model as a quality/processability indicator. Newly fabricated cohorts store the actual blended MFI achieved in the allocation step. No independent use-phase or EoU degradation factor is applied; changes in MFI arise from recycling transformations and subsequent blending.

Three selectable MFI mixing rules are implemented as structural alternatives:

$$\text{Arithmetic:} \quad \text{MFImix} = \Sigma i \; wi \; \text{MFIi}$$

$$\text{Geometric:} \quad \ln(\text{MFImix}) = \Sigma i \; wi \; \ln(\text{MFIi})$$

$$\text{Square-root:} \quad \sqrt{\text{MFImix}} = \Sigma i \; wi \; \sqrt{\text{MFIi}}$$

The reference setting is **Square-root** mixing. Stewart et al. [8] provide experimental evidence that non-arithmetic mixing behavior can occur in polyolefin blends; accordingly, the three rules are treated as structural sensitivity cases rather than interchangeable empirical truths.

Two application-compatibility modes are implemented. In published_range mode, the final fabricated blend must fall within the complete literature-derived MFI interval for the destination application. The conservative_centered_window mode uses the intersection of the literature range with [0.8 M̄g, 1.25 M̄g], where M̄g is the representative virgin application MFI. The 0.8 and 1.25 multipliers are user-defined modeling choices rather than universal HDPE processing limits.

The model is currently using Square-root and conservative_centered_window. The other methods are built into the model but not used in generating the results.

S6. Collection, base MRF, and sorting-resolution scenarios

EoU available for processing is divided into collected and uncollected fractions according to the recycling scenario. Uncollected material is routed to landfill and incineration while preserving the normalized 2015 residual split. All collected material passes through the base MRF. Base-MRF yield and burdens therefore apply to the entire collected mass.

Only MRF-recovered material actually required for downstream recycling receives additional optical sorting. Optical yield is applied once to the selected downstream feed, while the number of optical burden units depends on sorting resolution. This separates the physical mass-yield event from the additional burden associated with creating more resolved output streams. Sorting rejects are routed according to the technology loss routing; the current reference setting sends sorting losses to landfill. S4 uses the APR definition for Natural HDPE, Color HDPE, and Bulk/injection molding bales [9].

Table S 5 summarizes to which stream each application is sorted in each sorting scenario.

Table S 5. Application-to-sorting-stream map.

Application_ID	S1	S2	S3	S4	S5
Blow	MIX	LOW	LOW	NAT/COLOR	Blow
Fiber	MIX	HIGH	MID	REST	Fiber
Film	MIX	HIGH	HIGH	REST	Film
Injection	MIX	HIGH	HIGH	INJ	Injection
Pipe	MIX	LOW	LOW	REST	Pipe
Raffia	MIX	HIGH	MID	REST	Raffia
Rotomold	MIX	HIGH	HIGH	REST	Rotomold
Sheet	MIX	HIGH	MID	REST	Sheet
Tube_Profile	MIX	LOW	LOW	REST	Tube_Profile
Wire_Cable	MIX	HIGH	LOW	REST	Wire_Cable

For S4, Blow is split 50/50 between NAT and COLOR; therefore both stream IDs appear for that application.

S7. Technology yields, quality transformation, and burdens

Technology burdens are defined per tonne of process input. For a process r with input mass Mr,t and burden intensity br, the annual burden is:

$$\mathrm{Br,t} = \mathrm{Mr,t(input)}\ \mathrm{br}$$

To produce a required quantity of recycled output MRC through a recycling process with yield yr, the required recycling feed is MRC/yr. Thus, yield losses increase both material requirements and process burdens.

The MFI transformation for recycled output is represented with a technology-level multiplicative factor Kr:

$$\mathrm{MFIpost,r} = \mathrm{Kr}\ \mathrm{MFIpre}$$

The deterministic mechanical and dissolution factors are 0.74 and 0.94, respectively, based on recycled-to-virgin MFI quality ratios reported by Uekert et al. [10]. These factors are model proxies for technology-level quality response, not universal kinetic degradation laws. The same factor distribution is applied at each recycling cycle; the assumption is interpreted as a repeated-cycle process approximation after reblending with virgin resin, rather than full restoration of molecular or mechanical properties.

Table S 6 summarizes the technology performance metrics, economic burdens, the sources, and assumptions associated with each sorting or material recovery technology.

Table S 6. Technology yields, quality factors, energy, emissions, and costs.

Parameter	Basic MRF	MFI sorting	Mechanical recycling	Dissolution recycling	Virgin HDPE
Process yield	0.98	0.98	0.83	0.88	1.00
Yield standard deviation	0	0	0.06	0.07	0
Fraction of losses to landfill	1	1	1	1	0
Fraction of losses to incineration	0	0	0	0	0
MFI retention factor	1.00	1.00	0.74	0.94	1.00
MFI factor standard deviation	0	0	0.28	0.04	0
Primary energy (MJ/t input)	22.5	8.47	4,100	86,800	73,000
Emissions (kg CO_2/t input)	2.3	1.1	280	2,400	1,270
Cost (USD/t input)	22.4	10	630	1,100	790
Cost standard deviation (USD/t input)	0	0	30	90	30

1. Basic MRF performance is based on the Pressley/Combs MRF data sources. [11,12]
2. MFI-resolved sorting is an idealized model scenario rather than an established commercial technology for direct MFI sorting. Its energy intensity is estimated from an optical sorting unit operating at 40 kW rated power, 50% load, 10 Mg/h throughput, and 85% utilization, yielding 8.47 MJ/t. Sorting cost and emissions remain modeling assumptions pending source verification.
3. Mechanical recycling, dissolution recycling, and virgin HDPE parameters are based on Uekert et al. (2023) [10].
4. MFI retention factors represent the modeled changes in MFI associated with recycling processes, not complete descriptions of polymer quality. The values were extracted from the means and standard deviations collected from literature by Uekert et al. 2023 [10].
5. Loss fractions describe the allocation of process losses between disposal routes, not the total fraction of incoming material lost.

S8. Annual optimization and allocation

For each simulation year t the model solves a linear programming problem to determine the maximum quantity of quality-compatible recycled HDPE that can be incorporated into domestic applications. The optimization is solved independently for each year using SciPy's linprog implementation with the HiGHS solver. Therefore, the optimization is annual and myopic; decisions in year t do not anticipate future material availability or demand.

Let s denote a sorted HDPE stream, r a recycling technology, and g an application grade. The principal decision variable is

$$x_{s,r,g,t} \geq 0$$

where $x_{s,r,g,t}$ is the mass of recycled resin produced from sorted stream s, processed through recycling technology r, and allocated to application g in year t.

For compactness, the year subscript is omitted below.

S8.1 Recycled-resin allocation objective

For the standard quality-constrained allocation problem, the primary objective is to maximize the total mass of recycled resin incorporated into domestic fabrication:

$$\boxed{\max_{x} \; Z_{\mathrm{RC}} = \sum_{s}\sum_{r}\sum_{g} x_{s,r,g}}$$

Because annual product demand is fixed before the optimization is solved, maximizing recycled-resin use is equivalent to minimizing the amount of virgin resin required to satisfy that demand:

$$V_g = D_g - \sum_{s}\sum_{r} x_{s,r,g}$$

where D_g is total resin demand for application g.

Thus,

$$\max \sum_{s,r,g} x_{s,r,g}$$

is equivalent, for fixed demand, to

$$\min \sum_{g} V_g.$$

S8.2 Application-demand constraints

The recycled resin allocated to an application cannot exceed its annual resin demand:

$$\boxed{\sum_{s}\sum_{r} x_{s,r,g} \leq D_g \qquad \forall g}$$

The remaining application demand is satisfied with virgin resin:

$$V_g = D_g - \sum_s \sum_r x_{s,r,g}\,, \qquad V_g \geq 0.$$

This constraint prevents the model from producing or allocating recycled resin for which there is no domestic application demand.

S8.3 Sorted-feed availability and process-yield constraints

Let

- A_s = available mass of sorted stream s before downstream recycling,
- η_s = yield of the downstream sorting step,
- η_r = recycled-resin yield of recycling technology r.

To produce $x_{s,r,g}$ units of recycled resin, the corresponding upstream feed requirement is

$$\frac{x_{s,r,g}}{\eta_s \eta_r}.$$

Therefore, total use of each sorted stream cannot exceed its available mass:

$$\boxed{\sum_r \sum_g \frac{x_{s,r,g}}{\eta_s \eta_r} \leq A_s \qquad \forall s}$$

If $A_s^{\text{post-sort}}$ is defined after optical-sorting yield has already been applied, the constraint reduces to

$$\boxed{\sum_r \sum_g \frac{x_{s,r,g}}{\eta_r} \leq A_s^{\text{post-sort}}}$$

which is the equivalent form used when the sorted-stream inventory already represents recovered material.

These constraints ensure that recycled-resin allocation cannot exceed the physical mass available after sorting and recycling losses.

S8.4 MFI-compatibility constraints

Recycled material can be incorporated into an application only if the final recycled–virgin blend satisfies that application's allowable MFI range.

The supported mixing rules are expressed using the transformed coordinate

$$\phi(\text{MFI}) = \begin{cases} \text{MFI}, & \text{arithmetic mixing} \\ \ln(\text{MFI})\,, & \text{geometric mixing} \\ \sqrt{\text{MFI}}, & \text{square-root mixing.} \end{cases}$$

Let

$$\phi_{s,r} = \phi\left(\text{MFI}_{s,r}\right)$$

be the transformed MFI of recycled output from pathway (s, r), and let

$$\phi_g^V = \phi\left(\mathrm{MFI}_g^V\right)$$

be the transformed MFI of virgin resin for application g.

For application demand D_g, the transformed composition of the final recycled–virgin blend is

$$\phi_g^{\mathrm{blend}} = \frac{\sum_{s,r} x_{s,r,g}\,\phi_{s,r} + \left(D_g - \sum_{s,r} x_{s,r,g}\right)\phi_g^V}{D_g}.$$

The blend must remain between the transformed lower and upper application limits:

$$\phi_g^L \le \phi_g^{\mathrm{blend}} \le \phi_g^U.$$

Because blending is linear in transformed space, these become linear inequalities.

The upper MFI constraint is

$$\boxed{\sum_{s,r} x_{s,r,g}\left(\phi_{s,r} - \phi_g^V\right) \le D_g\left(\phi_g^U - \phi_g^V\right)}$$

and the lower MFI constraint is

$$\boxed{\sum_{s,r} x_{s,r,g}\left(\phi_{s,r} - \phi_g^V\right) \ge D_g\left(\phi_g^L - \phi_g^V\right)}$$

for every application g.

These constraints are what allow the arithmetic, geometric, and square-root mixing rules to use the same linear-programming framework.

S8.5 Non-negativity constraints

All pathway allocations are constrained to be non-negative:

$$\boxed{x_{s,r,g} \ge 0 \qquad \forall s, r, g}$$

No negative recycled-resin production, recycling feed, or application allocation is permitted.

S8.6 Fixed recycling-technology scenarios

For scenarios R2–R4, the recycling technology is either prescribed or constrained by a fixed feed split.

For a scenario using only mechanical recycling,

$$q_{\mathrm{Diss}} = 0$$

where q_r is total feed sent to recycling technology r.

For a scenario using only dissolution recycling,

$$q_{\mathrm{MR}} = 0.$$

For R4, in which 70% of recycling feed is assigned to mechanical recycling and 30% to dissolution recycling,

$$\boxed{q_{\mathrm{MR}} = 0.70\, q_{\mathrm{tot}}} \quad \boxed{q_{\mathrm{Diss}} = 0.30\, q_{\mathrm{tot}}}$$

with

$$q_{\mathrm{tot}} = q_{\mathrm{MR}} + q_{\mathrm{Diss}}.$$

An equivalent single linear equality is

$$\boxed{0.30\, q_{\mathrm{MR}} - 0.70\, q_{\mathrm{Diss}} = 0.}$$

The feed quantities can be related to recycled output through the technology yields:

$$q_r = \sum_{s,g} \frac{x_{s,r,g}}{\eta_r}.$$

Thus, the prescribed technology split is imposed on **recycling feed**, before technology-specific yield losses are applied.

Summary of objectives and constraints for R1-R4:

Table S 7. Linear-programming objective and constraints for recycled-resin allocation

Type	Mathematical formulation	Meaning
Objective — maximize recycled content / recycled resin use	$\max_x \sum_{s,r,g} x_{s,r,g}$	Maximize the total mass of recycled resin incorporated into domestic HDPE applications in the current year. Because annual demand is fixed, this is equivalent to minimizing the virgin resin required to satisfy that demand.
Application demand constraint	$\sum_{s,r} x_{s,r,g} \le D_g \quad \forall g$	Recycled resin allocated to application g cannot exceed the annual resin demand D_g for that application. Any remaining demand is met with virgin resin.
Sorted-feed availability / yield constraint	$\sum_{r,g} \frac{x_{s,r,g}}{\eta_s \eta_r} \le A_s \quad \forall s$	The upstream feed needed to produce the allocated recycled resin cannot exceed the available mass of sorted stream s. The sorting yield η_s and recycling yield η_r account for process losses.
Upper MFI compatibility constraint	$\sum_{s,r} x_{s,r,g}\,(\phi_{s,r} - \phi_g^V) \le D_g(\phi_g^U - \phi_g^V) \quad \forall g$	Ensures that the recycled–virgin blend allocated to application g does not exceed the upper allowable MFI limit for that application.
Lower MFI compatibility constraint	$\sum_{s,r} x_{s,r,g}\,(\phi_{s,r} - \phi_g^V) \ge D_g(\phi_g^L - \phi_g^V) \quad \forall g$	Ensures that the recycled–virgin blend does not fall below the lower allowable MFI limit for application g.
Non-negativity constraint	$x_{s,r,g} \ge 0 \quad \forall s, r, g$	Prevents negative recycled-resin production or allocation.
R1/R2 recycling-path constraint	$q_{\mathrm{Diss}} = 0$	All recycling feed is routed to mechanical recycling. R1 and R2 differ primarily in their collection rate, not the downstream recycling technology.
R3 recycling-path constraint	$q_{\mathrm{MR}} = 0$	All recycling feed is routed to dissolution recycling.
R4 recycling-share constraint	$q_{\mathrm{MR}} = 0.70 q_{\mathrm{tot}}$, $q_{\mathrm{Diss}} = 0.30 q_{\mathrm{tot}}$	Fixes the recycling-feed split at 70% mechanical recycling and 30% dissolution recycling before technology-specific yield losses are applied.
Equivalent R4 equality	$0.30 q_{\mathrm{MR}} - 0.70 q_{\mathrm{Diss}} = 0$	Linear equality form of the fixed 70/30 recycling-feed split, suitable for implementation in the LP solver.

S8.7 R5 recycling-technology allocation

R5 differs from the fixed recycling scenarios because both mechanical recycling and dissolution recycling are available simultaneously, and the model is allowed to determine how much sorted HDPE should be sent to each technology.

The same decision variables

$$x_{s,r,g}$$

therefore determine both:

1. which application receives the recycled resin, and
2. which recycling technology processes each sorted stream.

R5 uses a lexicographic sequence of objectives.

Stage 1: maximize recycled-resin utilization

The first-stage objective is

$$\boxed{\max_x Z_1 = \sum_s \sum_r \sum_g x_{s,r,g}}$$

subject to all demand, feed-availability, yield, MFI-compatibility, and non-negativity constraints defined above.

Let the optimal recycled-resin utilization from Stage 1 be

$$Z_1^*.$$

This stage determines the maximum quantity of quality-feasible recycled resin that can be incorporated into domestic fabrication in that year.

Stage 2: minimize annual allocation-dependent system cost

Among solutions that retain the same maximum recycled-resin utilization, R5 next minimizes annual allocation-dependent cost:

$$\boxed{\min_x Z_2 = C_{\text{alloc}}(x)}$$

where

$$C_{\text{alloc}} = C_{\text{sort}} + C_{\text{MR}} + C_{\text{Diss}} + C_{\text{virgin}}.$$

More explicitly,

$$C_{\text{alloc}} = \sum_s c_s\, q_s + \sum_r c_r\, q_r + c_V V,$$

where c_s, c_r, and c_V are the unit costs of downstream sorting, recycling technology r, and virgin resin production, respectively.

Because recycled resin displaces virgin production,

$$V = \sum_{g} \left(D_g - \sum_{s,r} x_{s,r,g} \right).$$

Base-MRF cost is not part of the allocation decision because all collected material passes through the base MRF regardless of the mechanical/dissolution allocation. It therefore contributes a constant term and does not affect the optimizer's choice between recycling pathways.

To preserve the Stage-1 optimum, the Stage-2 optimization is subject to

$$\boxed{\sum_{s,r,g} x_{s,r,g} \geq Z_1^* - \epsilon_1}$$

where ϵ_1 is the allowed lexicographic tolerance.

Using the configured relative and absolute tolerances,

$$\epsilon_1 = \max(\epsilon_{\text{abs}}, \epsilon_{\text{rel}} |Z_1^*|).$$

For the current max_rc objective,

$$\epsilon_{\text{rel}} = 10^{-9}, \qquad \epsilon_{\text{abs}} = 10^{-7}.$$

Therefore, cost minimization is allowed to choose a different recycling-technology allocation only if recycled-resin utilization remains effectively equal to the maximum identified in Stage 1.

Optional Stage 3: minimize recycling feed

A third lexicographic objective is implemented but disabled in the reported simulations. If enabled, it minimizes the total upstream recycling feed required to achieve the retained recycled-resin optimum:

$$\boxed{\min_x Z_3 = \sum_r q_r}$$

subject to retention of both the Stage-1 recycled-resin optimum and the Stage-2 cost optimum within their specified tolerances.

Because different recycling technologies have different yields, this objective favors solutions that achieve the required recycled-resin output with less total recycling feed.

Compact mathematical statement of the R5 problem

The R5 optimization can therefore be written conceptually as

$$\boxed{\text{lexmin} \left[-\sum_{s,r,g} x_{s,r,g}\,,\ C_{\text{alloc}}(x),\ \sum_r q_r \right]}$$

subject to

$$\sum_{s,r} x_{s,r,g} \leq D_g \qquad \forall g$$

$$\sum_{r,g} \frac{x_{s,r,g}}{\eta_s \eta_r} \leq A_s \qquad \forall s$$

$$\sum_{s,r} x_{s,r,g}\left(\phi_{s,r} - \phi_g^V\right) \le D_g\left(\phi_g^U - \phi_g^V\right) \qquad \forall g$$

$$\sum_{s,r} x_{s,r,g}\left(\phi_{s,r} - \phi_g^V\right) \ge D_g\left(\phi_g^L - \phi_g^V\right) \qquad \forall g$$

$$x_{s,r,g} \ge 0$$

The third objective is included in the implementation but is disabled by default for the reported scenarios.

Priority	Objective	Mathematical objective	Relative tolerance	Absolute tolerance	Interpretation
1	max_rc	$\max \sum_{s,r,g} x_{s,r,g}$	10^{-9}	10^{-7}	Maximizes quality-feasible recycled resin incorporated into domestic fabrication; equivalent to minimizing virgin-resin requirement for fixed annual demand.
2	min_cost	$\min C_{\mathrm{alloc}}(x)$	10^{-9}	10^{-4}	Selects the least-cost allocation among solutions that retain the Stage-1 recycled-resin optimum.
3	min_feed	$\min \sum_{r} q_r$	configurable	configurable	Optional objective that minimizes required recycling feed after retaining the higher-priority optima; disabled in the reported simulations.

S9. Key assumptions summary

Table S 8. MFI-resolved DMFA model key assumptions summary

Assumption	Treatment	Status / rationale
Historical quality	Historical cohorts initialized at virgin application MFI	Historical recycled-content quality unavailable
Film vs sheet mass split	50/50	Author assumption
Pipe vs tube/profile mass split	50/50	Author assumption
Residual commercial 'Other'	Redistributed proportionally to explicit grades	Avoids unsupported single MFI
Pipe MFI	Open interval 0–0.1; deterministic 0.05	Operationalization of literature <0.1
Fiber MFI uncertainty	0.9–1.1 around approximately 1.0	±10% author uncertainty
S4 natural/color bottle split	50/50	Author assumption
S4 injection mapping	All Injection to injection-bulky-rigid bale	Upper-bound mapping approximation
Repeated-cycle MFI response	Same technology-factor distribution each cycle	Process approximation after reblending
Modern sector transition	Abrupt switch in 2019	Structural assumption; sensitivity candidate
Collected-feed inventory	0 t	Unused MRF-recovered material is landfilled
Conservative compatibility	Published range ∩ [0.8M̃,1.25M̃]	Optional model-defined conservative case

Appendix A. Complete reference data tables

Table S 9. Historical U.S. HDPE demand and reference stock/scrap series.

Year	US_Production_MMt	US_Consumption_MMt	Reference_Stock_MMt	Reference_Scrap_MMt
1950	0.063333	0.0584921	0	0
1951	0.0745094	0.0688142	0.0227443	0.0357341
1952	0.0707839	0.0653735	0.0481363	0.0434059
1953	0.0894113	0.082577	0.0685066	0.0449878
1954	0.100588	0.0928991	0.0915418	0.0595224
1955	0.130391	0.120425	0.116269	0.0681498
1956	0.149019	0.137628	0.150007	0.0866589
1957	0.171372	0.158273	0.188019	0.0995838
1958	0.182548	0.168595	0.230849	0.115406
1959	0.234705	0.216765	0.274034	0.125369
1960	0.268234	0.247731	0.33248	0.158268
1961	0.298038	0.275257	0.399115	0.181037
1962	0.35392	0.299735	0.471272	0.203036
1963	0.406076	0.347984	0.548875	0.22206
1964	0.484311	0.416258	0.638848	0.257927
1965	0.55882	0.472144	0.748375	0.306633
1966	0.655683	0.559917	0.877038	0.343366
1967	0.733917	0.632221	1.03005	0.406769
1968	0.879211	0.756583	1.19745	0.46467
1969	1.04313	0.913394	1.39887	0.554984
1970	1.11764	0.985777	1.6434	0.668646
1971	1.2294	1.08479	1.89607	0.732874
1972	1.41568	1.25491	2.16759	0.81301
1973	1.63921	1.43031	2.4833	0.938903
1974	1.67646	1.43993	2.85165	1.06162
1975	1.49019	1.29788	3.20453	1.0867
1976	1.75097	1.52009	3.47048	1.03162
1977	1.89999	1.671	3.8038	1.18641
1978	2.04901	1.85108	4.18642	1.28799
1979	2.27254	2.00098	4.61541	1.42166
1980	1.99811	1.85331	5.08369	1.53223
1981	2.12965	1.96044	5.45563	1.48094
1982	2.23534	1.8915	5.84158	1.57404
1983	2.58144	2.20208	6.17466	1.558
1984	2.76016	2.41329	6.62233	1.7539
1985	3.02597	2.5555	7.14136	1.89372
1986	3.25776	2.72963	7.68305	2.01324
1987	3.62653	2.98153	8.25811	2.15396
1988	3.81024	3.16503	8.89039	2.34857
1989	3.67507	3.03035	9.55282	2.50189
1990	3.78166	3.25675	10.1368	2.44569
1991	4.17902	3.17863	10.7663	2.62655
1992	4.44891	3.40936	11.3151	2.62911
1993	4.50924	3.56591	11.9531	2.77059
1994	5.04267	3.84411	12.6366	2.88155
1995	5.08531	3.86612	13.368	3.11182
1996	5.61239	4.1991	14.0731	3.16021
1997	5.70493	4.39673	14.874	3.39717
1998	5.86233	4.54575	15.7516	3.51813
1999	6.28871	5.04037	16.641	3.65536
2000	6.33588	4.95116	17.7638	3.91645
2001	6.93282	5.1529	18.8047	3.90906
2002	5.74049	4.92532	19.8146	4.14186
2003	5.72772	5.08148	20.8485	3.89034
2004	6.31552	5.48393	21.8398	4.08899
2005	5.81299	5.50184	22.9842	4.33824
2006	6.33404	5.72617	24.0763	4.40848
2007	6.5524	5.54164	25.1575	4.6436
2008	4.34885	4.2355	26.0627	4.6352
2009	4.80188	4.09321	26.4567	3.84038
2010	4.78782	4.22911	26.7014	3.84753
2011	4.79881	4.39259	26.9891	3.94036
2012	5.07676	4.69423	27.4108	3.96982
2013	4.8758	4.78465	27.9284	4.17548
2014	4.8685	4.99216	28.4545	4.25739
2015	5.16566	5.05484	29.0074	4.43803
2016	5.11392	5.11001	29.5685	4.49257
2017	5.24647	5.36736	30.1637	4.51357
2018	6.95816	6.35238	30.8612	4.66863

Table S 10. Full application-sector joint allocation matrix.

Period	Application_ID	Sector_ID	Joint_Share_Total_HDPE	Sector_Share_Within_Application
historical_1950_2018	Injection	Packaging	0.17331	0.651647
historical_1950_2018	Injection	Transportation	0.017664	0.0664167
historical_1950_2018	Injection	Building_Construction	0.0178602	0.0671543
historical_1950_2018	Injection	Electrical_Electronic	0.00607305	0.0228347
historical_1950_2018	Injection	Consumer_Institutional	0.038067	0.143132
historical_1950_2018	Injection	Industrial_Machinery	0.00136312	0.00512532
historical_1950_2018	Injection	Other	0.0116197	0.0436899
historical_1950_2018	Film	Packaging	0.165807	0.779291
historical_1950_2018	Film	Building_Construction	0.0227825	0.107078
historical_1950_2018	Film	Consumer_Institutional	0.0161862	0.0760749
historical_1950_2018	Film	Industrial_Machinery	0.0005796	0.00272412
historical_1950_2018	Film	Other	0.00741104	0.0348319
historical_1950_2018	Sheet	Packaging	0.130252	0.612186
historical_1950_2018	Sheet	Transportation	0.00885032	0.0415965
historical_1950_2018	Sheet	Building_Construction	0.0402688	0.189263
historical_1950_2018	Sheet	Electrical_Electronic	0.00456423	0.0214519
historical_1950_2018	Sheet	Consumer_Institutional	0.019073	0.089643
historical_1950_2018	Sheet	Industrial_Machinery	0.00102446	0.00481496
historical_1950_2018	Sheet	Other	0.00873281	0.0410442
historical_1950_2018	Blow	Packaging	0.0818184	0.854547
historical_1950_2018	Blow	Transportation	0.00185312	0.0193548
historical_1950_2018	Blow	Consumer_Institutional	0.00798716	0.0834215
historical_1950_2018	Blow	Industrial_Machinery	0.000429011	0.00448078
historical_1950_2018	Blow	Other	0.00365703	0.0381956
historical_1950_2018	Pipe	Building_Construction	0.0567178	0.888579
historical_1950_2018	Pipe	Industrial_Machinery	0.000961955	0.0150706
historical_1950_2018	Pipe	Other	0.00615001	0.0963501
historical_1950_2018	Tube_Profile	Transportation	0.0104288	0.163385
historical_1950_2018	Tube_Profile	Building_Construction	0.047451	0.743399
historical_1950_2018	Tube_Profile	Industrial_Machinery	0.000804785	0.0126083
historical_1950_2018	Tube_Profile	Other	0.00514519	0.0806079
historical_1950_2018	Fiber	Transportation	0.00512362	0.16054
historical_1950_2018	Fiber	Building_Construction	0.00777078	0.243485
historical_1950_2018	Fiber	Consumer_Institutional	0.0110417	0.345974
historical_1950_2018	Fiber	Industrial_Machinery	0.000395386	0.0123888
historical_1950_2018	Fiber	Other	0.00758339	0.237613
historical_1950_2018	Wire_Cable	Transportation	0.00316582	0.148793
historical_1950_2018	Wire_Cable	Building_Construction	0.00960294	0.451338
historical_1950_2018	Wire_Cable	Electrical_Electronic	0.00163266	0.0767349
historical_1950_2018	Wire_Cable	Consumer_Institutional	0.00454836	0.213773
historical_1950_2018	Wire_Cable	Industrial_Machinery	0.000244304	0.0114823
historical_1950_2018	Wire_Cable	Other	0.00208252	0.0978786
historical_1950_2018	Raffia	Packaging	0.0193644	0.910125
historical_1950_2018	Raffia	Consumer_Institutional	0.000945181	0.0444235
historical_1950_2018	Raffia	Industrial_Machinery	0.000101536	0.0047722
historical_1950_2018	Raffia	Other	0.000865526	0.0406797
historical_1950_2018	Rotomold	Transportation	0.00199403	0.187439
historical_1950_2018	Rotomold	Consumer_Institutional	0.00644589	0.605914
historical_1950_2018	Rotomold	Industrial_Machinery	0.000230817	0.0216968
historical_1950_2018	Rotomold	Other	0.00196756	0.18495
modern_2019_plus	Injection	Packaging	0.0548082	0.206079
modern_2019_plus	Injection	Building_Construction	0.0153805	0.0578306
modern_2019_plus	Injection	Automotive	0.0556179	0.209123
modern_2019_plus	Injection	Consumer_Goods	0.050819	0.19108
modern_2019_plus	Injection	Industrial	0.0568969	0.213932
modern_2019_plus	Injection	Other	0.0324349	0.121955
modern_2019_plus	Film	Packaging	0.080527	0.378477
modern_2019_plus	Film	Building_Construction	0.0301304	0.141613
modern_2019_plus	Film	Consumer_Goods	0.0331848	0.155969
modern_2019_plus	Film	Industrial	0.0371537	0.174622
modern_2019_plus	Film	Other	0.03177	0.149319
modern_2019_plus	Sheet	Packaging	0.0446384	0.209801
modern_2019_plus	Sheet	Building_Construction	0.0375798	0.176625
modern_2019_plus	Sheet	Automotive	0.0301986	0.141933
modern_2019_plus	Sheet	Consumer_Goods	0.027593	0.129687
modern_2019_plus	Sheet	Industrial	0.0463396	0.217796
modern_2019_plus	Sheet	Other	0.0264166	0.124158
modern_2019_plus	Blow	Packaging	0.035146	0.36708
modern_2019_plus	Blow	Automotive	0.00792559	0.0827784
modern_2019_plus	Blow	Consumer_Goods	0.0144835	0.151272
modern_2019_plus	Blow	Industrial	0.0243236	0.254046
modern_2019_plus	Blow	Other	0.013866	0.144823
modern_2019_plus	Pipe	Building_Construction	0.0293668	0.460079
modern_2019_plus	Pipe	Industrial	0.0241414	0.378215
modern_2019_plus	Pipe	Other	0.0103216	0.161705
modern_2019_plus	Tube_Profile	Building_Construction	0.0214401	0.335895
modern_2019_plus	Tube_Profile	Automotive	0.0172289	0.26992
modern_2019_plus	Tube_Profile	Industrial	0.0176252	0.276127
modern_2019_plus	Tube_Profile	Other	0.00753561	0.118058
modern_2019_plus	Fiber	Building_Construction	0.00283866	0.0889446
modern_2019_plus	Fiber	Automotive	0.00684331	0.214424
modern_2019_plus	Fiber	Consumer_Goods	0.00625285	0.195923
modern_2019_plus	Fiber	Industrial	0.00700068	0.219355
modern_2019_plus	Fiber	Other	0.0089794	0.281354
modern_2019_plus	Wire_Cable	Building_Construction	0.00436383	0.2051
modern_2019_plus	Wire_Cable	Automotive	0.00526006	0.247223
modern_2019_plus	Wire_Cable	Consumer_Goods	0.00320414	0.150595
modern_2019_plus	Wire_Cable	Industrial	0.00538103	0.252908
modern_2019_plus	Wire_Cable	Other	0.00306754	0.144174
modern_2019_plus	Raffia	Packaging	0.00928036	0.436177
modern_2019_plus	Raffia	Consumer_Goods	0.0019122	0.0898734
modern_2019_plus	Raffia	Industrial	0.00642269	0.301866
modern_2019_plus	Raffia	Other	0.00366135	0.172083

Period	Application_ID	Sector_ID	Joint_Share_Total_HDPE	Sector_Share_Within_Application
modern_2019_plus	Rotomold	Automotive	0.00222566	0.209212
modern_2019_plus	Rotomold	Consumer_Goods	0.00305044	0.286741
modern_2019_plus	Rotomold	Industrial	0.00341527	0.321035
modern_2019_plus	Rotomold	Other	0.00194693	0.183011

Joint shares are IPF-derived allocation coefficients constrained to the application and sector marginals; they are not directly observed market shares.

Table S 11. Historical prior (seed) matrix, 1950–2018 for the joint application x use-sector shares

Application	Packaging	Transportation	Building & Construction	Electrical & Electronic	Consumer & Institutional	Industrial Machinery	Other	Justification of seed values
Injection	2	3	1	2	3	3	2	Packaging: common rigid-packaging use but not exclusive. Transportation: strong use of injection-molded automotive parts. Building: possible but secondary. E&E: plausible housings/components. Consumer: strong household/consumer-goods use. Industrial: strong molded industrial components. Other: broadly plausible residual applications.
Film	3	0	2	0	2	2	2	Packaging: strong support from bags/films. Transportation and E&E: not treated as meaningful HDPE film destinations in the prior. Building: construction films are plausible. Consumer, industrial and other: secondary uses such as liners, films and protective products.
Sheet	2	2	3	2	2	3	2	Packaging: plausible sheet/rigid-packaging use. Transportation: secondary sheet applications. Building: strong use in panels/linings/construction products. E&E: plausible sheet/housing applications. Consumer: secondary household goods. Industrial: strong fabricated-sheet use. Other: residual plausible.
Blow molding	3	1	0	0	2	3	2	Packaging: strong bottle/container use. Transportation: limited reservoirs/tanks. Building and E&E: no clear matching category. Consumer: plausible household containers. Industrial: strong drums, tanks and industrial containers. Other: residual plausible.
Pipe	0	0	3	0	0	2	1	Building: directly supported by pipes/ducts. Industrial: process piping is plausible. Other: small residual use. Packaging, transportation, E&E and consumer sectors are structurally excluded in this prior.
Tube/Profile	0	2	3	0	0	2	1	Building: strong support from profiles, ducts and construction extrusions. Transportation: plausible automotive profiles/tubing. Industrial: secondary extrusion applications. Other: residual. Packaging, E&E and consumer are excluded.
Fiber	0	2	1	0	2	2	3	Other: strongest support from textiles/fiber-type product categories. Transportation, consumer and industrial: plausible secondary fiber uses. Building: minor plausible use. Packaging and E&E excluded.
Wire/Cable	0	3	3	3	2	3	2	Transportation, building, E&E and industrial all have strong cable/conduit/insulation-type relevance. Consumer: plausible appliance/electrical-goods use. Other: residual. Packaging excluded.
Raffia	3	0	0	0	1	3	2	Packaging: strong woven sacks/strapping/raffia use. Industrial: strong bulk-bag/industrial woven-product use. Consumer: weak residual possibility. Other: plausible. Transportation, building and E&E excluded.
Rotomolding	0	2	0	0	3	3	2	Consumer: strong large hollow goods/recreational products. Industrial: strong tanks and containers. Transportation: plausible tanks/components. Other: residual. Packaging, building and E&E excluded.

Table S 12. Modern prior matrix, 2019 onward prior (seed) matrix for the joint application x use-sector shares

Application	Packaging	Building & Construction	Automotive	Consumer Goods	Industrial	Other	Justification of seed values
Injection	2	1	3	3	3	2	Packaging: plausible rigid packs/closures. Building: minor molded products. Automotive: strong molded-component use. Consumer: strong household goods. Industrial: strong molded industrial parts. Other: broadly plausible.
Film	3	2	0	2	2	2	Packaging: strong bags/film use. Building: construction film is plausible. Automotive: excluded as a meaningful HDPE-film destination. Consumer, industrial and other: secondary film/liner applications.
Sheet	2	3	2	2	3	2	Building: strong sheet/lining/panel use. Industrial: strong fabricated-sheet applications. Automotive: secondary sheet use. Packaging, consumer and other: plausible but less dominant.
Blow molding	3	0	1	2	3	2	Packaging: strong bottles/containers. Automotive: limited tanks/reservoirs. Consumer: household containers. Industrial: strong drums/tanks. Building excluded; other residual.
Pipe	0	3	0	0	2	1	Building: directly supported pipe/duct use. Industrial: process piping plausible. Other: small residual. Packaging, automotive and consumer excluded.
Tube/Profile	0	3	2	0	2	1	Building: strong profiles/ducts. Automotive: plausible extruded profiles/tubing. Industrial: secondary profile use. Other: residual. Packaging and consumer excluded.
Fiber	0	1	2	2	2	3	Other: strongest textile/fiber grouping. Automotive: plausible textile/fiber components. Consumer and industrial: plausible. Building: weak secondary. Packaging excluded.
Wire/Cable	0	3	3	2	3	2	Building, automotive and industrial: strong cable/conduit use. Consumer: plausible appliance/electrical product use. Other: residual. Packaging excluded.
Raffia	3	0	0	1	3	2	Packaging: strong woven sacks/strapping. Industrial: strong bulk bags/industrial woven products. Consumer: weak residual possibility. Other: plausible. Building and automotive excluded.
Rotomolding	0	0	2	3	3	2	Consumer: strong recreational/household hollow products. Industrial: strong tanks/containers. Automotive: plausible tanks/components. Other: residual. Packaging and building excluded.

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